

8.1 Confidence Interval on the Mean of a Normal Distribution (Variance Known)

We look again at the ten measurements from the **uniformly distributed** random variable X on the integers $a = 1 \leq x \leq b = \text{unknown}$. See Section 7.2. We call $\hat{b} = 479$ a **point estimate** of b .

However, we can also determine the unknown b using an **interval estimate**, e.g. “in 95% of the cases b is between 353.3 and 747.5”.

We call an interval like $[353.3; 747.5]$ a **95%-confidence interval for b** . See Figure 8-1 in the book or Figure 8.1 in this document. In Figure 8.1 a confidence interval for μ (the true mean is zero!) is constructed 20 times. Each interval is determined using 20 measurements (= blue dots). In one case (i.e. 5%) the real value of μ does not belong to the interval.

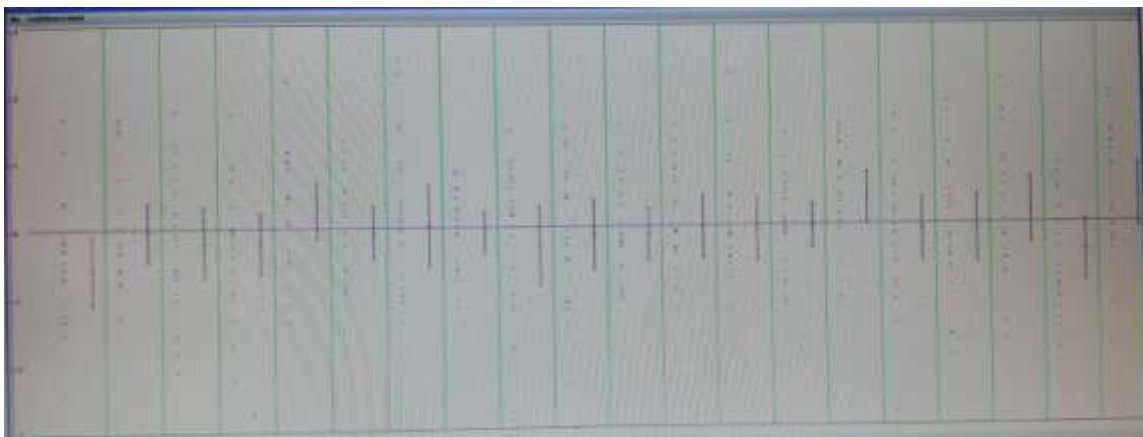


Figure 8.1 Repeated construction of a confidence interval for μ

Definition

The interval $[\vartheta_1; \vartheta_2]$ is called a **100*(1- α)-confidence interval (CI) for ϑ** , if

$$P(\Theta_1 = L(X_1, X_2, \dots, X_n) \leq \vartheta \leq U(X_1, X_2, \dots, X_n) = \Theta_2) = 1 - \alpha$$

The quantity $1 - \alpha$ is called the **confidence coefficient** (often: $\alpha = 0.05$ or $\alpha = 0.01$).

Theorem

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution with unknown mean μ and known variance σ^2 . Then a **100(1 - α)% CI** for μ is given by

$$[\vartheta_1; \vartheta_2] = \left[\bar{x} - z_{1/2\alpha} \cdot \frac{\sigma}{\sqrt{n}}; \bar{x} + z_{1/2\alpha} \cdot \frac{\sigma}{\sqrt{n}} \right]$$

where $z_{1/2\alpha}$ satisfies $\Phi(z_{1/2\alpha}) = 1 - 1/2\alpha$, i.e. $z_{1/2\alpha} = \text{invNorm}(1 - 1/2\alpha)$.

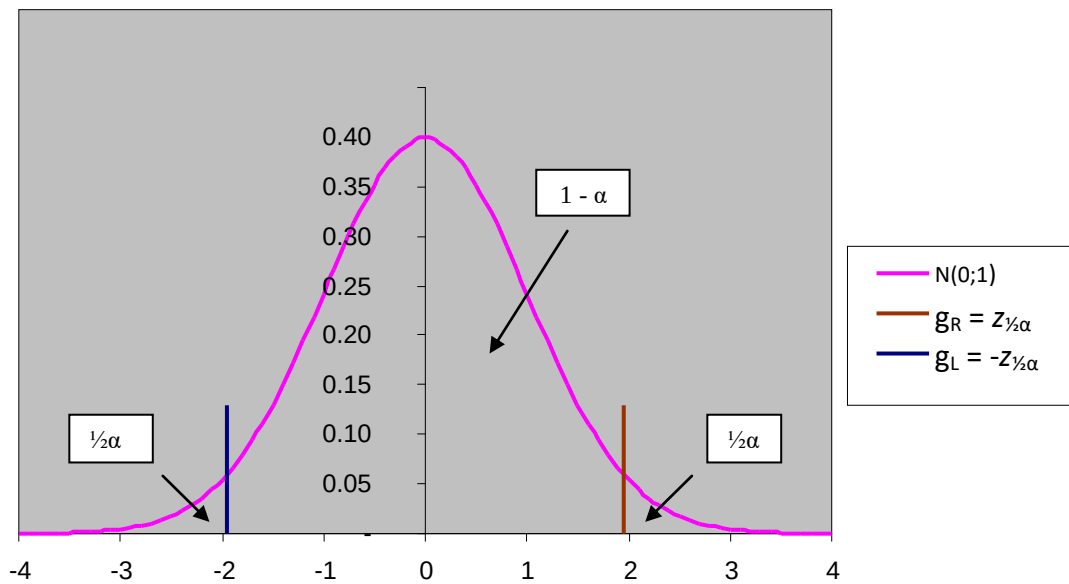


Figure 8.2 Standard normal probability density function

Proof

Since $\bar{X} \sim N(\text{mean} = \mu; \text{variance} = \sigma^2/n)$, it follows

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0; 1) \quad \text{standard normal distribution}$$

So

$$P(-z_{\alpha/2} \leq Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}) = 1 - \alpha \quad \Leftrightarrow$$

$$\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Examples

1. Determine a **95%-CI** for μ , if $X \sim N(\mu = \text{unknown}; \sigma^2 = 9)$, $\bar{x} = 5$ and $n = 100$.

Since $\alpha = 0.05$, $z_{\alpha/2}$ is given by

$$z_{0.025} = \text{invNorm}(0.975, 0, 1) = 1.9600 \quad \text{gc/distr/3}$$

Substitution gives

$$[\vartheta_1; \vartheta_2] = [5 - 1.9600 \cdot \frac{3}{\sqrt{100}}; 5 + 1.9600 \cdot \frac{3}{\sqrt{100}}] = [4.4120; 5.5880]$$

NB Using the option **ZInterval** of the **GC** gives [4.412; 5.588] gc/stat/tests/7

- Determine the **minimum sample size** n , if $X \sim N(\mu = \text{unknown}; \sigma^2 = 4)$ and if the length of the 99%-CI for μ is at most 0.2.

Since $\alpha = 0.01$, it follows that $z_{\frac{1}{2}\alpha} = z_{0.005} = \text{invNorm}(0.995, 0, 1) = 2.5758$.

The length of the CI is equal to $\vartheta_2 - \vartheta_1$.

Since

$$\vartheta_2 - \vartheta_1 = \bar{x} + z_{\frac{1}{2}\alpha} \cdot \frac{\sigma}{\sqrt{n}} - (\bar{x} - z_{\frac{1}{2}\alpha} \cdot \frac{\sigma}{\sqrt{n}}) = 2 \cdot z_{\frac{1}{2}\alpha} \cdot \frac{\sigma}{\sqrt{n}}$$

we must solve the inequality below

$$2 \cdot z_{\frac{1}{2}\alpha} \cdot \frac{\sigma}{\sqrt{n}} = 2 \cdot 2.5758 \cdot \frac{2}{\sqrt{n}} = \frac{10.3033}{\sqrt{n}} \leq 0.2$$

So

$$\sqrt{n} \geq 51.5166 \quad \Leftrightarrow \quad n \geq 2653.9601$$

Therefore

$$n = 2654$$

Remarks

- The quantity $(\vartheta_2 - \vartheta_1)/2$ is called the **precision** of the estimation. So in the last example the difference between $\bar{x}$ and μ is 0.1 or less.

It is easy to verify that the following formula applies to a sample size n and a **maximum permitted difference E** between $\bar{x}$ and μ :

$$n \geq \left(\frac{z_{\frac{1}{2}\alpha} \cdot \sigma}{E} \right)^2$$

- The **larger** the **confidence coefficient** $1 - \alpha$, the **longer** the confidence interval. So a smaller value for α leads to a less clear statement regarding the unknown parameter.
- The **larger** the **sample size** n , the **shorter** the confidence interval. More measurements therefore lead to more accurate estimates.
- In the case of a **large sample size** n , the requirement $X \sim N(\text{mean} = \mu; \text{variance} = \sigma^2)$ is canceled, because it follows from the **CLS** that $\bar{X} \approx N(\text{mean} = \mu; \text{variance} = \frac{\sigma^2}{n})$, regardless of the distribution of X . Also σ need not be known, because $s \approx \sigma$.
- It is also possible to define a **one-sided confidence bound**:

$$\begin{array}{llll} \text{a } 100(1 - \alpha)\% \text{ upper-confidence bound:} & \mu & \leq & \bar{x} + z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}} \\ \text{a } 100(1 - \alpha)\% \text{ lower-confidence bound:} & \mu & \geq & \bar{x} - z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}} \end{array}$$